

RESPONSE TO REQUEST FOR INFORMATION

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Docket No. HHS-ONC-2026-0001

Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care

Submitted by

PHAROS HEALTH

San Francisco, California

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Submitted to:

Assistant Secretary for Technology Policy (ASTP)
U.S. Department of Health and Human Services

Dr. Thomas Keane, MD, MBA.
Assistant Secretary for Technology Policy

Mr. Steve Posnack
Principal Deputy National Coordinator

Re: HHS-ONC-2026-0001 | Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care

Dear Sirs,

We appreciate the opportunity to provide concrete, experience-based feedback on how the Department of Health and Human Services (HHS) can accelerate the adoption of Artificial Intelligence (AI) across the healthcare continuum. As an AI vendor with deep experience in healthcare data analytics, clinical NLP, and real-world AI deployment, we offer this response grounded in operational reality.

While significant attention is appropriately being paid to AI applications at the point of care such as ambient documentation, clinical decision support, and diagnostic assistance, our response focuses on a complementary domain: interoperability and surveillance, where AI can transform quality improvement, reporting, patient safety monitoring, health information exchange and data normalization.

Our response details 10 specific use cases across burden reduction, patient safety, interoperability, and quality improvement. These use cases emerged from hundreds of conversations with clinicians, quality leaders, health plan executives, and health IT professionals. Our team combines clinical expertise with years of production AI and traditional software deployment in healthcare settings. We present only applications where we have observed genuine operational need and where we are actively developing solutions. This is not theoretical speculation; it reflects what healthcare organizations are asking for and what we are building.

We look forward to partnering with HHS to move these concepts from pilot to policy.

Kind regards

Felix Brann
Chief Executive Officer
Pharos Health, San Francisco CA

Overview

For more than two decades, federal agencies and standards development organizations have worked to make clinical information computable. The goal has been straightforward: when data can be read by machines, organizations can automate processes that would otherwise require someone to open a chart, read through documentation, and manually record what they find. This manual work, known as chart abstraction, remains one of the largest drivers of administrative cost in American healthcare. Quality measure reporting alone costs U.S. physician practices an estimated \$15.4 billion annually, much of it spent on exactly this kind of review.

The EHR Incentive Program, HL7 v2, C-CDA, and FHIR are examples of successive efforts to reduce this burden by encouraging digitizing health data and establishing common data formats. These standards have enabled significant automation, yet chart abstraction persists because clinicians document much of what they know in narrative form. A physician's assessment, a nurse's observations, a social worker's notes: this information often appears in free text rather than structured fields. Furthermore there continues to be weak adoption and implementation of standardization of data across the healthcare continuum.

AI-based clinical natural language processing offers a way to extend existing interoperability infrastructure into narrative documentation and weakly standardized data. These methods extract structured meaning from clinical notes, discharge summaries, and legacy records without requiring clinicians to change how they document. The result is programmable access to information that was previously available only through human review.

This response describes use cases where AI-enabled processing works alongside existing standards to reduce administrative burden, support patient safety surveillance, and strengthen health information infrastructure. The examples reflect operational deployments within health plans, health systems, and healthcare analytics organizations. Each use case connects directly to ASTP's long standing priorities: making clinical data computable, reducing the cost of manual abstraction, and enabling the interoperability infrastructure that supports a learning health system.

Use Case Summary

Clinical Registry Reporting: Hospitals submit detailed patient data to national quality databases, but extracting this information requires staff to manually read through charts for hours per case. AI reads operative notes, pathology reports, and clinical documentation to extract registry variables automatically, reducing review time by 70-90%.

Clinical Quality Measurement and Reporting: Healthcare quality measurement depends on standardized, structured data, but implementation of data standardization continues to lag across the industry. Extending measurement into the free-text clinical notes where much of the relevant information lives remains prohibitively expensive, affordable only to the most resource-intensive measurement programs that can fund manual chart review. Emerging AI technology enables less dependency on implementation of standards and extension of measurement into the free text.

Guideline Adherence Monitoring: Hospitalized patients often fail to receive recommended preventive treatments like blood clot prophylaxis, but gaps remain invisible until harm occurs. AI monitors patients in real-time against clinical guidelines and can identify trends where adherence to clinical practice guidelines is responsible for harm and additional costs.

Utilization Management: Determining whether a hospital admission is medically necessary requires case managers to hunt through records finding scattered clinical data points. AI pre-populates medical necessity reviews by extracting vital signs, lab values, and clinical findings from across the chart, letting reviewers focus on judgment rather than data gathering.

Incidental Findings: Unexpected medical observations discovered during diagnostic tests or procedures conducted for unrelated clinical reasons. Because these findings often reside in unstructured narrative notes rather than standardized codes, they frequently lack systematic follow-up or administrative visibility.

Elimination of Prior Authorization: Combined implementation of AI powered Guideline Adherence Monitoring and Utilization Management enable a "gold card" qualification where trusted providers receive automatic approval rather than per-request review.

Population Health Gap Closure: Patients appear overdue for preventive care when their screenings were done at outside facilities and documented only in scanned PDFs. AI extracts procedure dates and findings from narrative documents to identify which gaps are real and which were already addressed elsewhere.

Safety Event Detection: Most patient harm goes undetected because voluntary safety reporting captures only 10-20% of adverse events. AI scans all patient records for clinical warning signs.

Root Cause Analysis: When serious safety events occur, investigators must reconstruct what happened from fragmented notes written by different clinicians. AI synthesizes documentation from all sources into a unified timeline, revealing communication gaps and missed warning signs.

Sensitive Data Segmentation: Privacy laws require protecting certain health information before sharing records, but code-based filters miss sensitive content documented in narrative text. AI understands the meaning of clinical documentation and identifies protected information regardless of how it was recorded.

Clinical Registry Reporting

Understanding of Need

Hospitals and physician practices voluntarily participate in clinical registries to benchmark their performance against national standards and identify opportunities for improvement. A registry is essentially a database that collects standardized information about patients with specific conditions or procedures, stroke patients, surgical cases, cancer diagnoses, cardiac interventions. By comparing their outcomes to aggregated data from hundreds of other institutions, healthcare organizations can see where they excel and where they fall short.

The challenge is that registries require detailed clinical information that must be manually extracted from medical records. A trained abstractor, typically a nurse or health information specialist, reads through operative notes, progress notes, laboratory results, and discharge summaries to find specific data elements and enter them into the registry system. This process is called chart abstraction, and it is extraordinarily time-consuming.

A typical health system participates in dozens of registries across clinical domains: ACS NSQIP alone collects 135 variables per case. A routine case takes an experienced abstractor about 60 minutes; complex cases with multiple complications can require 5-6 hours of chart review. A health system performing 22,000 surgical cases annually dedicates over 11,000 labor hours, more than five full-time employees, to this single registry.

DOMAIN	EXAMPLE REGISTRIES
Surgical	ACS TQIP, ACS NSQIP, ACS MBSAQIP, STS ACSD
Cardiac & Thoracic	ACC CathPCI, ACC/SVS VQI, ACC LAAO, ACC Chest Pain-MI, GWTG Resuscitation
Oncology	NAACCR-NPCR, ACS NCDB, NAACCR-SEER, ASCO QOPI
Neurology	GWTG Stroke, AANS N2QOD, AANS SRS

Current barriers:

- Registry variables reside in operative notes and clinical documentation, not structured fields
- Determining complications requires clinical judgment applied to narrative text
- EHR implementations vary in how clinicians document the same concepts

Proposed Approach

AI-based natural language processing can read clinical documentation and extract registry variables programmatically. Rather than replacing human abstractors, these systems accelerate their work by identifying relevant information across hundreds of pages of medical records and presenting findings with links to source documentation for human verification. The AI accommodates the variability in how different clinicians and different EHR systems document the same clinical concepts. When information is ambiguous or conflicting across sources, the system flags the discrepancy for human

review rather than making assumptions. Production deployments demonstrate 95% agreement with expert human abstractors while reducing time per case by as much as 90%.

Regulatory Alignment

Registry participation supports CMS quality reporting requirements and accreditation standards. AI-assisted abstraction reduces the burden on participating practices without requiring changes to how clinicians document care, aligning with federal goals for administrative simplification.

Complexity of “Last Known Well”

Scenario: For the Get With The Guidelines-Stroke registry from the AHA, the abstractor must determine the "Last Known Well" time, the last moment the patient was observed to be neurologically normal. This single data element is critical because it determines eligibility for clot-dissolving medication (tPA), which must be administered within hours of symptom onset.

Clinical Context:

The abstractor finds conflicting time references across multiple documentation sources: Data validation audits show "Last Known Well" achieves only 85.3% abstractor agreement, the lowest reliability of any variable in the registry. The complexity arises because abstractors must determine which source is most reliable (direct witness accounts take precedence over second-hand reports) and reconcile conflicting information.

Pt: Male, 72 | Encounter: ED Stroke Alert

EMS run sheet: Patient found on floor at 14:32, daughter states 'he was fine at breakfast'

ED triage note (14:58): Last known well per EMS: 08:00

Neurology note (15:20): Spoke with daughter, patient watched morning news, she left for work at 10:15

Nursing note (15:05): Family states patient took medications at 9 AM

Code Stroke form: field left blank

AI Application: The AI extracts all time references from EMS documentation, nursing notes, and physician notes. It identifies the daughter's direct witness account in the neurology note as the most reliable source according to registry abstraction hierarchy guidelines. It determines Last Known Well as 10:15 and presents a complete timeline showing all documented times with links to source documentation, allowing the abstractor to verify the determination in minutes rather than reconstructing the timeline manually.

Measurable Impact:

- Addresses the registry variable with lowest inter-rater reliability (85.3%)
- Synthesizes 4+ documentation sources in seconds vs. 15-20 minutes manual review
- Reduces tPA eligibility determination errors from conflicting timestamps

Clinical Quality Measurement and Reporting

At its core, healthcare quality measurement is an abstraction problem. Clinical research identifies the markers of high-quality, evidence-based care; quality measurement then seeks to abstract those markers from the primary medical record to evaluate performance. For decades, this abstraction was a manual, labor-intensive process performed by human reviewers. The transition to electronic Clinical Quality Measures (eCQMs) is designed to automate this, reducing the cost and effort of abstraction by standardizing both the data format such as CCDA and FHIR and the measure logic expression with the Clinical Quality Language or CQL.

The Limitations of Structural Standardization

Standardizing clinical data, moving from HL7 v2 to C-CDA and now toward FHIR has resulted in a matrix of non-uniformity where even when organizations adopt a shared standard, local implementation choices, custom data mappings, and varying rates of adoption across the continuum of care leave the data "messy."

As a result, a Chief Quality Officer at an academic medical center cannot simply "run" a measure across their enterprise and trust the result. They find that even standardized fields for common labs or vitals remain unreliable until they undergo resource-intensive, organization-specific reconciliation. In this environment, the administrative data still fails to speak the language of the clinician, remaining an abstraction of a proxy rather than an abstraction of the care itself.

The Clinical Narrative and Free Text

The second, more significant challenge is that the most vital parts of the clinical story are never "coded." Critical indicators of clinical judgment, patient nuance, and medical necessity reside in unstructured clinical notes and free text. Because these narratives are not standardized, they remain invisible to eCQM developers and the rigid logic of CQL.

The move toward Digital Quality Measures (dQMs) is intended to expand the data ecosystem to include registries and devices, but it does not resolve this fundamental visibility gap. dQMs still largely rely on Clinical Quality Language (CQL), a machine-readable expression language that, while powerful, is restricted to structured inputs. Consequently, the industry faces a compounding barrier: the logic standard lacks broad adoption, and the clinical narrative remains unreadable to the machine.

Proposed Approach

AI-based natural language processing can normalize poorly standardized data and extend the reach of electronic quality measurement by extracting measure-relevant data elements from clinical documentation.

AI functions as a dynamic abstraction service that can interpret "messy" standardized data and unstructured narrative text with equal high-fidelity, no longer requiring every health system to perfectly map every data element to a rigid schema before a measure can function.

AI can also expand the range of clinical quality measures that can be implemented because the complexity of data standardization and chart abstraction is no longer a barrier to implementation.

Regulatory Alignment

This approach supports ASTP requirements for electronic quality measurement while acknowledging the practical reality that not all clinical information exists in structured form. It extends the reach of CQL-based quality measurement to the unstructured content that currently requires manual abstraction. AI enabled abstraction allows for more complex and relevant measures to be implemented that can focus incentive payments directly to high quality care.

The SEP-1 Challenge

Scenario: The SEP-1 measure (Severe Sepsis and Septic Shock Management Bundle) is one of the most complex and time-consuming quality measures in healthcare. It requires determining "time zero", the moment when a patient first met criteria for severe sepsis, and then verifying that a bundle of treatments (blood cultures, antibiotics, fluids, repeat lactate) was completed within 3 and 6 hours of that moment.

Clinical Context:

Consider a patient who arrives at the emergency department. Determining "time zero" requires clinical judgment: when did this patient first meet the clinical definition of severe sepsis? Was it at triage when infection was suspected? At 14:45 when vital signs showed organ dysfunction? Studies show that abstractors agree on time zero only 36% of the time, meaning nearly two-thirds of cases involve legitimate disagreement about measure interpretation.

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Pt: Male, 58 | Encounter: ED Sepsis Alert

Triage note at14:32: fever, cough, confusion
Vital signs at14:45: temp 39.1°C, BP 88/54, RR 24
Lactate at15:18: 4.2 mmol/L
Antibiotics administered at16:15
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AI Application: The AI reviews all documentation from the encounter, identifies the earliest evidence of suspected infection combined with evidence of organ dysfunction, and determines time zero as 14:45 (when vital signs demonstrated hypotension meeting SIRS criteria in a patient with suspected infection). It then calculates whether each bundle element was completed within the required timeframe, presenting a timeline with linked source evidence for each determination.

Expanding Measure Development: The CAP-Pharos Collaboration

When abstraction costs fall, entirely new categories of quality measurement become feasible. Pharos is currently collaborating with the College of American Pathologists (CAP) to apply neurosymbolic AI to

pathology quality measures, work presented at the Council of Medical Specialty Societies (CMSS) Annual Meeting in 2025.

The Problem: CAP's Qualified Clinical Data Registry (QCDR) measures require data elements that live exclusively in narrative pathology reports: histologic tumor grade, margin status, lymphovascular invasion, depth of invasion. Consider what a real pathology report looks like:

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CPT Code(s) : 88305x4
SPECIMEN SOURCE:
RIGHT FOREHEAD / RIGHT ANTERIOR NECK LINE / LEFT MID STERNUM / LEFT FOREARM
FINAL PATHOLOGIC DIAGNOSIS:
A. Skin, right forehead, shave biopsy: Actinic keratosis.
B. Skin, right anterior neck line, shave biopsy: Basal cell carcinoma,
nodular type; - present at peripheral and deep margins.
C. Skin, left mid sternum, shave biopsy: Basal cell carcinoma, superficial
type; - present at peripheral margin.
D. Skin, left forearm, shave biopsy: Squamous cell carcinoma in situ; -
present at peripheral and deep margins.
This case has been reviewed and dictated by [Name], M.D., Ph.D., Board Certified
Dermatopathologist.
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The yellow-highlighted elements are precisely what the quality measure needs to capture, but they exist only in narrative text. No structured code or billing field contains this information. Existing keyword and rule-based extraction tools proved inadequate: a rule searching for “GE* biops*” to find gastric biopsies would match on a patient named George who had an ear biopsy. CAP was forced to abandon automated collection of clinically important measures, including a gastric intestinal metaplasia measure, because no existing technology could reliably extract the required data elements from narrative pathology text.

See Exhibit 1 - Neurosymbolic AI for automated clinical quality measure abstraction in pathology - CMSS 25

The Approach: Rather than prompting a general-purpose large language model, the system converts each measure specification into a neurosymbolic decision tree: a structured pipeline that decomposes the measure into discrete sub-problems. This approach constrains AI tightly to the precise definitions laid out in the measure specifications, minimizing hallucination and producing fully auditable reasoning and evidence citations for every determination.

The Results: The system was evaluated across four CAP QCDR measures on 2,000 pathology reports (500 randomly subsampled per measure from a feed of 100,000 reports). All reports were independently double-abstracted by experienced third-party abstractors trained on each measure, establishing a human inter-rater reliability (IRR) baseline. Final gold-standard labels were determined by third-party RN abstraction managers. The neurosymbolic models were then evaluated against the same gold standard.

	IRR	Accuracy
Human	0.918	95.4%
AI	0.922	95.7%

No statistical difference between human and AI abstractors.

The Implication: This collaboration demonstrates a broader principle directly relevant to this RFI. Over-simplifying quality measures to rely only on structured data fields, simply because that is what current tools can process, undermines the clinical purpose of measurement.

Clinically complex measures without added burden. Professional societies can develop measures that reflect current clinical practice guidelines without requiring costly manual abstraction or adding documentation burden to clinicians. Measure data can be collected directly from the narrative documentation clinicians already create. Furthermore measures could be developed that leverage patient interviews, sentiment analysis and other non-structured data sources to capture the patient’s “voice” in their care.

Reduced friction for participating practices. Practices no longer need to filter or pre-process data before submission to registries. Vastly more accurate measurement means dramatically fewer case corrections and resubmissions, reducing the operational cost of registry participation.

Accelerated measure development. New measures can be applied instantly to a large corpus of existing reports, with AI surfacing edge cases for review by measure developers. This allows measure refinement before deployment, compressing the development cycle from years to weeks. The CAP collaboration is an early proof point: measures that were previously intractable are now being automated at human-level accuracy, enabling quality measurement to expand into domains that were inaccessible to existing technology.

Guideline Adherence Monitoring

Clinical practice guidelines exist for nearly every dimension of patient care: when to initiate antibiotics, how to manage blood glucose, which prophylactic therapies hospitalized patients should receive, and how to respond to early signs of clinical deterioration. These recommendations reflect decades of accumulated evidence about what improves outcomes and prevents harm.

Medicine will always require clinical judgment and nuance. But guidelines define a baseline standard of care - specifying the interventions, timing, and thresholds most consistently associated with better outcomes. When care systematically deviates from that baseline, it can signal breakdowns in processes, communication, or oversight. Detecting those deviations early offers a powerful opportunity to identify emerging safety risks before they result in patient harm. And when avoidable harm does occur, a central question remains: was there a moment when care diverged from best practice?

Today, answering that question is extraordinarily labor-intensive. Determining whether care was guideline-concordant requires a trained clinician to manually review the chart - every note, order, lab result, and nursing assessment - reconstruct the clinical timeline, and compare it against what evidence-based recommendations dictate. This depth of review is typically reserved for retrospective root cause analyses and applied to only a small subset of cases. Performing the same analysis continuously, across an entire patient population, remains effectively intractable.

Complicating matters further, guidelines include exceptions. A patient may have a legitimate contraindication to a recommended treatment. A clinician may document sound reasoning for an alternative approach. Distinguishing a process failure from an appropriate clinical decision requires reading and interpreting the documentation.

Finally, clinical practice guidelines are not static. New evidence emerges continuously, yet formal guideline updates often lag behind the science by months or years. Conversely, some recommendations persist long after their evidentiary basis has weakened or been superseded. Once new guidance is issued, there is rarely a systematic mechanism to evaluate how rapidly it is adopted, how faithfully it is implemented, or whether it produces the intended outcomes in real-world settings. This creates a feedback gap: guidelines shape care, but care data are not routinely used to inform the refinement, retirement, or recalibration of the guidelines themselves.

Proposed Approach

AI can continuously monitor whether actual care delivery aligns with guideline expectations across the full scope of clinical practice. For each patient, the system maps their clinical situation against relevant guideline pathways, identifying what should happen based on their conditions and risk factors, checking whether those care steps occurred, and reading clinical documentation to determine whether any deviation has documented clinical justification. AI does more than just identify guideline misses, it can reason about which process failures actually drive harm and provide targeted intervention opportunities to the quality team.

This transforms quality monitoring from expensive retrospective chart abstraction on a small sampling of patients into prospective surveillance at scale, on every patient for every care transaction.. Furthermore, rather than discovering guideline gaps only after harm occurs, organizations can identify deviations while there is still time to intervene. AI can surface to Care teams only high value alerts when a patient's care appears to diverge from guidelines without documented rationale, enabling correction before complications develop.

Regulatory Alignment

This approach supports CMS quality programs by shifting from measuring harm after the fact to preventing it prospectively. It enables organizations to demonstrate systematic guideline adherence rather than relying on outcome-based measures that capture only the cases where process failures resulted in detectable harm. Furthermore, it leads to tighter feedback loops - new processes can be implemented and then immediately evaluated and tuned to specific patient populations.

Missed VTE Protocol

Scenario: Orthopedic surgery patients are at particularly high risk for VTE due to the nature of the procedure and post-operative immobility. Hospital protocol and national guidelines require VTE prophylaxis for all orthopedic surgery patients unless specifically contraindicated.

Clinical Context:

The admission nursing assessment correctly identifies the patient as high-risk for VTE based on the planned procedure. Pre-operative orders are placed by the surgical team, including antibiotics, pain management, and post-operative physical therapy, but no VTE prophylaxis order is included in the order set. On post-operative day 1, the patient begins ambulating with physical therapy as documented in the PT notes. By post-operative day 2, the medication administration record still shows no prophylaxis has been given. A review of all clinical documentation, nursing notes, physician progress notes, pharmacy notes, reveals no documented contraindication to anticoagulation and no documentation that the patient declined mechanical prophylaxis. The omission appears to be an inadvertent failure to order a standard component of post-operative care.

TIME	CLINICAL NOTE	GUIDELINE COMPLIANCE
Day 0 14:00	Admission nursing assessment completed. VTE risk score calculated: HIGH (orthopedic procedure, age 68, BMI 32). Risk documented in chart.	✓ Joint Commission NPSG.03.05.01 — VTE risk assessment required for all surgical admissions.
Day 0 16:00	Pre-op orders placed: Cefazolin 2g IV, Morphine PCA, PT consult. No VTE prophylaxis ordered. No contraindication documented.	✗ ACCP / AAOS — VTE prophylaxis required for all major orthopedic surgery patients unless specifically contraindicated.
Day 0 18:30	Total knee replacement completed without complication. Standard post-op order set activated. VTE prophylaxis not included in order set.	✗ WHO Surgical Safety Checklist — VTE prophylaxis plan must be verified before incision. Order set gap allowed omission.
Day 1 06:00	AM nursing assessment. Patient ambulating with walker. MAR: Cefazolin, Morphine, Ondansetron. No anticoagulant administered. No contraindication on file.	✗ ACCP — Chemical and mechanical prophylaxis should begin within 12–24 hours post-operatively for high-risk patients.
Day 1 15:00	Patient reports increasing left calf pain and swelling. Positive Homan's sign. Stat duplex ultrasound ordered.	✗ Symptoms consistent with developing DVT. Prophylaxis window missed. No protocol triggered reassessment.

Day 2 08:00	Duplex ultrasound confirms acute DVT, left popliteal vein. Heparin drip initiated. Discharge delayed. Vascular surgery consult.	⚠️ AVOIDABLE HARM — DVT secondary to missed prophylaxis. Guideline-compliant prophylaxis reduces post-surgical VTE risk by 50–60%.
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AI Application: On post-operative day 1, the AI system detects the combination of high VTE risk, absence of any prophylaxis order, and no documented contraindication. It generates an alert to the charge nurse and the clinical pharmacist, identifying the specific gap: high-risk orthopedic surgery patient without prophylaxis, no contraindication documented. The pharmacist contacts the surgical team, an order is placed, and prophylaxis begins on post-operative day 1, before any harm occurs.

Utilization Management

Understanding of Need

When a physician decides that a patient needs to be admitted to the hospital, stay in the ICU, or receive a particular treatment, someone has to verify that the clinical situation justifies that level of care. This process is called utilization management, and it exists because healthcare resources are expensive and sometimes overused. A patient who could safely recover at home shouldn't occupy a hospital bed.

Health plans and hospitals employ nurses and other clinical staff as case managers or utilization review specialists. These reviewers compare patient information against standardized medical necessity criteria, commercial products like InterQual or MCG that specify what clinical findings justify different levels of care.

Healthcare administrative costs reached an estimated \$274.5 billion in 2017, with utilization management representing a substantial component. Case managers spend hours each day on data extraction tasks that don't require clinical judgment, simply finding and copying information from one system to another.

The Utilization Management Criteria Landscape

Health plans rarely rely on a single criteria set. Utilization management operates through a patchwork of guideline products, each governing different clinical domains, layered with plan-specific customizations and subject to a strict hierarchy of authority.

Medical Necessity Criteria Sets in Use

CRITERIA SET	DOMAIN	TYPICAL USE
InterQual <i>Optum / Change Healthcare</i>	General acute, subacute, rehab, long-term care	Primary criteria for most large commercial and Medicare Advantage plans
MCG <i>Hearst Health</i>	General acute, ambulatory, behavioral health	Primary criteria for many payers; often used alongside InterQual
ASAM Criteria <i>American Society of Addiction Medicine</i>	Substance use disorder	Mandated by most states for SUD level-of-care determinations
LOCUS / CALOCUS-CASII	Behavioral health	Adults (LOCUS) and children/adolescents (CALOCUS-CASII) service intensity
NCCN Guidelines <i>National Comprehensive Cancer Network</i>	Oncology	Chemotherapy protocols, cancer-related imaging, treatment pathways
Hayes Technology Assessments	New / experimental procedures	Evidence sufficiency for emerging technologies and novel treatments
Carelon <i>formerly AIM / Beacon</i>	Radiology, cardiology, musculoskeletal	Delegated specialty UM using proprietary clinical guidelines

Apollo Managed Care	General acute	Lower-cost alternative used by smaller plans, IPAs, and TPAs
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UM Decision Hierarchy

Each review follows a defined hierarchy of authority. Federal mandates override all other sources.

	SOURCE	AUTHORITY	EXAMPLE
1	Federal / State Mandates	Legal requirement to be applied first	<i>NCDs, LCDs for Medicare; state-mandated SUD criteria</i>
2	Plan-Specific Medical Policies	Internal policy supersedes commercial criteria	<i>Bariatric Surgery Policy with plan-specific BMI thresholds</i>
3	Commercial Criteria	InterQual, MCG, NCCN, ASAM, etc.	<i>Standard level-of-care review for acute medical admission</i>
4	Physician Clinical Judgment	Escalation when no criteria fit	<i>Peer-to-peer review with Medical Director</i>

The AI Opportunity

This patchwork creates the core problem AI can solve. A single admission review may require the case manager to determine which criteria set applies, navigate the decision hierarchy, extract clinical data from the EHR, map it against the correct criteria, and document the determination, all within increasingly compressed turnaround requirements.

Proposed Approach

AI can extract clinical data from electronic health records and map it against medical necessity criteria automatically. Rather than requiring a case manager to search through the chart for each data point, the system identifies relevant clinical information, calculates any derived scores (like pneumonia severity indices), and presents a pre-populated review with links to source documentation.

This approach doesn't remove the case manager from the process, it removes the tedious data extraction that consumes most of their time.

Regulatory Alignment

Accurate level-of-care determination supports appropriate resource utilization without changing clinical workflows. The approach reduces administrative burden while maintaining the clinical oversight that utilization management requires.

Operational Example

Optum/Change Healthcare Webinar August 31, 2023

Scenario:

An 89-year-old patient presents to the emergency department with a three-day history of shortness of breath, cough, weakness, and fever. O2 saturation ranges from 85-87% on room air, improving to 93-97% on 3L nasal cannula. The emergency physician orders admission for community-acquired pneumonia.

Clinical Context:

The case manager needs to find specific clinical data points scattered across the medical record. She opens the patient's chart in one screen, the medical necessity criteria tool in another, and the hospital's submission portal in a third. She navigates to vital signs to find the O2 saturation. This process takes 15-25 minutes of pure data extraction before any clinical judgment is applied.

AI Application: When the admission order is placed, the AI automatically triggers a utilization review. It extracts vital signs from the nursing flowsheet, lab values from the laboratory system, and imaging results from radiology reports. It confirms that antibiotics were administered (a treatment criterion). It presents a pre-populated InterQual review showing each criterion, the patient's corresponding value, and links to the source documentation. The case manager reviews the populated form, verifies accuracy, and approves the admission in a fraction of the time manual review would require.

Incidental Findings

Understanding the Need: The Abstraction of Secondary Clinical Data

Clinical documentation is primarily structured to support the immediate reason for a patient encounter. However, diagnostic procedures frequently reveal incidental findings which are unexpected abnormalities that are unrelated to the primary complaint but represent significant early warning signs for chronic disease or malignancy. These findings, such as a lung nodule identified during a cardiac scan, often contain the "second story" of a patient's health. While these observations are clinically relevant, they are difficult to track systematically because they fall outside the immediate scope of the primary diagnosis and the resulting billing codes.

The Limitations of Billing-Centric Records

The central barrier to managing incidental findings is that current documentation systems are designed for targeted, episodic care. Because an incidental finding is secondary to the chief complaint, it is rarely assigned a discrete ICD-10 code during the initial encounter. Without a standardized code, the finding does not trigger administrative alerts or follow-up protocols. The data exists only within the narrative text of the clinical record. This gap creates a significant risk where the physician notes may contain actionable findings, yet a majority do not receive the recommended follow-up care because manual chart review is too resource-intensive to perform at scale.

The Challenges of Manual and Rules-Based Review

Traditional methods of identifying these findings rely on manual abstraction by nursing or administrative staff, which is often cost-prohibitive. Attempts to automate this process using rules-based logic or simple keyword searches frequently fail. These systems often produce high rates of false positives by flagging benign variations or miss subtle clinical nuances because they cannot interpret the context of the physician's prose.

Proposed Approach

AI-based natural language processing provides a mechanism to bridge this gap by performing high-fidelity abstraction across the entire clinical narrative. Unlike manual review, AI models can process vast quantities of unstructured text to identify specific, actionable findings and reconcile them with existing patient data. This technology allows organizations to move from reactive documentation to a proactive system of care. By automating the identification of incidental findings, AI ensures that secondary clinical data is captured and integrated into the patient's longitudinal record, enabling timely intervention and more accurate risk assessment without increasing the administrative burden on clinicians.

Elimination of Prior Authorization

Understanding of Need

Prior authorization is a requirement imposed by health insurance payers that physicians obtain approval before delivering certain medical services. Prior authorization exists because payers lack visibility into whether providers make appropriate clinical decisions. When a physician orders an expensive biologic medication or an advanced imaging study, the payer has no way of knowing whether that decision reflects sound clinical judgment or unnecessary utilization. Prior authorization fills this information gap by requiring providers to submit documentation proving medical necessity for each individual request ahead of time.

The time to respond to a prior authorization request can impact patient care and CMS recognized this burden in the CMS-0057-F Interoperability and Prior Authorization Final Rule, which requires payers to process standard requests within 72 hours and urgent requests within 24 hours. The rule acknowledges that authorization delays directly harm patients and impose unsustainable costs on providers.

The burden is significant. Physicians spend an average of 14 hours per week on prior authorization (American Medical Association, 2024). Practices complete 43 requests per physician per week (AMA 2023 Physician Survey, 2024). Staffing costs have increased 43% since 2019 (MGMA Regulatory Burden Report, 2024). Patients wait for treatments their physicians have already determined they need (AMA, 2024).

This is one area where we see cautionary misuse of AI as a way to augment process with AI on both sides of a prior authorization exchange - where two AI bots are battling each other and the winner is likely not going to be the patient.

But the underlying question prior authorization tries to answer is straightforward: does this provider make clinically appropriate decisions? If a payer could answer that question at the provider level rather than the request level, individual prior authorizations would become unnecessary.

This is the concept behind "gold carding" which is granting trusted providers automatic approval for services that would otherwise require prior authorization. Several states have enacted gold carding legislation, and CMS has encouraged payers to reduce prior authorization burden for providers with strong track records.

State Gold Carding Legislation

STATE	LEGISLATION	THRESHOLD	PERIOD	VOLUME	NOTES
Texas	HB 3459 (2021); HB 3812 (2025)	90%	12 Months	5 per service	First state to enact; updated 2025
Michigan	PA 60 (2022)	90%+	Annual	Payer-defined	Threshold varies by payer
Arkansas	Act 575 (2023)	90%	12 Months	N/A	6-month period for Rx drugs

STATE	LEGISLATION	THRESHOLD	PERIOD	VOLUME	NOTES
West Virginia	W. Va. Code §33-15-4s	90%	6 Consecutive Mos.	30 requests/yr	
Louisiana	La. R.S. 22:1020.61	92%	2 Years	10 requests/yr	Longest evaluation window
Wyoming	WY §26-55-112	90%	12 Months	5 per service	
Colorado	CO §10-16-112.5	80%*	12 Months	24 requests	*Lowest threshold enacted
Vermont	HB 960	100%	6 Months	30 requests/yr	Highest threshold required
New Mexico	13.10.31.12 NMAC	90%	Annual	N/A	Regulation-based, not statute

Proposed Approach

AI enables providers to demonstrate evidence-based care and sound decision-making directly from clinical documentation, creating an vehicle for gold card qualification that goes far beyond claims metrics. By applying the capabilities described in Utilization Management and Guideline Adherence Monitoring across a provider's patient panel, payers can systematically assess whether that provider's care aligns with medical necessity criteria and clinical guidelines.

Providers who demonstrate consistent alignment earn gold card status, the automatic approval for categories of services that would otherwise require prior authorization. The authorization burden shifts from per-request documentation to periodic provider-level validation.

This approach benefits both sides. Providers escape the administrative burden of individual authorizations. Payers reduce their own processing costs while maintaining confidence that approved services are medically appropriate. Patients receive timely care without authorization delays.

Regulatory Alignment

CMS has explicitly encouraged payers to reduce prior authorization burden for providers with demonstrated quality. State gold carding laws in Texas, Michigan, and elsewhere require payers to exempt qualifying providers from prior authorization requirements. AI-based provider evaluation creates the measurement infrastructure these policies require, systematic, auditable assessment of clinical decision-making that can support gold card qualification at scale.

Population Health Gap Closure

Understanding of Need

Value-based care contracts reward healthcare organizations for keeping patients healthy, not just for treating them when they're sick. A key component of this approach is identifying "care gaps", patients who are overdue for recommended preventive services or chronic disease management. Reporting can show which patients are overdue for services, conduct outreach campaigns encouraging patients to schedule appointments, and work to ensure that preventive care doesn't fall through the cracks.

The challenge is that care gap identification requires knowing the complete picture of a patient's healthcare, and no single organization has that complete picture. A patient might have their colonoscopy done by a gastroenterologist who uses a different EHR system or the record of a completed preventive service can appear in the free text of a clinical note or coded in a non-standard format that current reporting systems miss. Organizations will often manually review charts (or pay outside services) to find patients eligible for care gaps by reviewing poorly standardized data and free text.

Health information exchanges (HIEs) have emerged to address this fragmentation by aggregating data across healthcare organizations in a region. When a patient visits the emergency department, the HIE can pull together records from their primary care doctor, specialists, and previous hospital visits. This gives providers a more complete picture, but only if they can actually find and interpret the relevant information.

CANDIDATE CARE GAPS		HEDIS OR CMS MEASURE ALIGNMENT	
Preventive and Proactive Care			
Whether eligible patients received recommended preventive services or risk screenings within defined time intervals		Breast Cancer Screening HEDIS BCS; CMS125v14 Cervical Cancer Screening HEDIS CCS; CMS124v14 Colorectal Cancer Screening HEDIS COL; CMS130v14 Childhood Immunization Status HEDIS CIS; CMS117v14 Depression Screening CMS2v15 Falls Risk Screening CMS139v14 HIV Screening CMS349v8	
Timely assessment and follow-up after abnormal or incomplete preventive screening results		Documented Assessment After Mammogram HEDIS DBM-E Follow-Up After Abnormal Mammogram HEDIS FMA-E	
Care Coordination and Transitions			

Completion of required follow-up, coordination, or reconciliation activities after a qualifying clinical event	Transitions of Care HEDIS TRC Medication Reconciliation Post-Discharge HEDIS MRP Follow-Up After Hospitalization for Mental Illness HEDIS FUH Follow-Up After ED Visit for Mental Illness HEDIS FUM Follow-Up After High-Intensity SUD Care HEDIS FUI Closing the Referral Loop CMS50v14
Whether clinical information is transmitted back to the referring provider following specialty or external care	Closing the Referral Loop: Receipt of Specialist Report CMS50v14
Chronic and Longitudinal Care	
Evidence of ongoing monitoring or control for patients with chronic conditions across care settings	Controlling High Blood Pressure HEDIS CBP; CMS165v14 Diabetes Glycemic Control HEDIS GSD; CMS122v14 Kidney Health Evaluation HEDIS KED; CMS951v4 Eye Exam for Diabetes HEDIS EED; CMS131v14 Asthma Medication Ratio HEDIS AMR
Initiation and persistence of evidence-based therapies over defined treatment periods	Antidepressant Medication Management CMS128v14 Statin Therapy for ASCVD HEDIS SPC; CMS347v9 Statin Therapy for Diabetes HEDIS SPD Beta-Blocker Persistence After MI HEDIS PBH
Ongoing management following acute events requiring sustained therapy or rehabilitation	Cardiac Rehabilitation HEDIS CRE Pharmacotherapy Management of COPD Exacerbation HEDIS PCE
Safety and Appropriateness of Care	
Avoidance of unnecessary, low-value, or non-recommended services	Appropriate Testing for Pharyngitis CMS146v14 Appropriate Treatment for URI CMS154v14 Avoidance of Antibiotics for Bronchitis HEDIS AAB Imaging for Low Back Pain HEDIS LBP Non-Recommended PSA Screening HEDIS PSA
Exposure to high-risk medications or unsafe prescribing patterns	High-Risk Medications in Older Adults HEDIS DAE; CMS156v14 Drug-Disease Interactions HEDIS DDE Opioid High Dosage / Multiple Providers HEDIS HDO, UOP Deprescribing Benzodiazepines HEDIS DBO
Behavioral Health and Whole-Person Care	

Monitoring, follow-up, and continuity of care for behavioral health conditions	Depression Remission at Twelve Months CMS159v14 Follow-Up Care for Children Prescribed ADHD Medication HEDIS ADD-E; CMS136v15 Initiation and Engagement of SUD Treatment HEDIS IET; CMS137v14
Assessment of functional status, symptoms, or patient-reported outcomes	Functional Status After Hip Replacement CMS56v14 Heart Failure Functional Status CMS90v15 CAHPS / HOS Measures
Clinical Documentation Integrity / Completeness	
Whether required clinical information is documented, updated, or returned in a structured and timely manner to support continuity of care, quality reporting, and downstream clinical decision-making	Documentation of Current Medications in the Medical Record CMS68v15 Closing the Referral Loop: Receipt of Specialist Report CMS50v14 Transitions of Care: Receipt of Discharge Information HEDIS TRC

Proposed Approach

AI-based natural language processing can extract clinical evidence from narrative documentation and scanned documents flowing through health information exchanges. When structured data doesn't capture completed care, the AI reads consultation notes, procedure reports, and other documentation to identify the information that would close care gaps.

The system processes HIE data across multiple contributing organizations, looking for evidence that recommended services have been completed.

Regulatory Alignment

This approach supports value based pay models and helps organizations keep their populations healthy.

Cancer Screenings

Scenario: The care management team at an ACO is preparing an outreach campaign to encourage patients to schedule colonoscopies.

Clinical Context:

The ACO's EHR shows no colonoscopy on file for this patient. However, an HIE query reveals that the patient had an emergency department visit at an outside hospital six months ago. Among the documents returned is a scanned PDF of a gastroenterology consultation note from that visit. The note states: "Prior colonoscopy 3/15/2023 at [outside facility], no polyps, recommended surveillance interval 10 years. Patient does not require repeat screening at this time." The patient is not actually overdue for colorectal cancer

screening, he had a normal colonoscopy less than two years ago and doesn't need another one for eight more years.

AI Application: The AI processes the scanned PDF from the HIE query. It identifies the reference to colonoscopy, extracts the date (March 15, 2023), the findings (no polyps), and the recommended surveillance interval (10 years). It determines that this patient's colorectal cancer screening gap should be closed. The patient is removed from the outreach list, and the quality measure is calculated correctly.

HEDIS 2026

<https://www.ncqa.org/report-cards/health-plans/state-of-health-care-quality-report/colorectal-cancer-screening-col/>

Safety Event Detection

Understanding of Need

Voluntary incident reporting, the primary mechanism hospitals use to identify patient harm, captures only 10 to 20 percent of actual adverse events as many of the safety events are not visible to staff but are buried in the clinical documentation.

To address this gap, the Institute for Healthcare Improvement developed the Global Trigger Tool (Griffin & Resar, 2009). In addition to voluntary reports, trained reviewers examine patient records for "triggers," clinical findings statistically associated with adverse events: a creatinine value that doubles from baseline, administration of an opioid reversal agent, an unexpected ICU transfer. Each trigger flags a case for clinical review to determine whether harm occurred. The triggers can be automated with current reporting and technology but they only identify that an event "might" have occurred and require manual review by a trained clinician to determine if a safety event occurred.

The methodology is effective but constrained by manual effort. IHI recommends reviewing 10 randomly selected records every two weeks. At that rate, the vast majority of discharges are never examined, and the events within them are never counted.

Proposed Approach

AI can execute trigger tool logic against all patient records, not just a random sample. The system scans clinical data, laboratory values, medication administrations, vital signs, and clinical documentation, looking for the same trigger conditions that human reviewers would identify. When triggers fire, the AI can then inspect the entire record of a patient identified by an IHI trigger to determine if a safety event has occurred avoiding the clinician review step current IHI trigger methodology requires.

Regulatory Alignment

This approach supports Joint Commission patient safety requirements by enabling more comprehensive surveillance. It allows organizations to identify systemic issues without the proportional staffing increase that manual comprehensive review would require.

IHI Trigger Tool Study

Scenario: In 2011, researchers applied the IHI Global Trigger Tool to 795 patient records at three major academic medical centers, comparing it against voluntary incident reporting and AHRQ Patient Safety Indicators using the same patient sample. Voluntary reporting identified 4 adverse events. AHRQ indicators identified 35. The trigger tool identified 354. Adverse events were present in one-third of all admissions.

The trigger tool works by scanning clinical data for specific signals, called triggers, that are statistically associated with harm. The IHI defines triggers across six categories:

CATEGORY	TRIGGER
Medication (M)	Antidote administration, abrupt stops, creatinine doubling → adverse drug events, overdoses, nephrotoxicity
Care (C):	Unexpected ICU transfer, blood transfusion, code/arrest → clinical deterioration, missed warning signs
Intensive Care (I)	Re-intubation, new dialysis in ICU → ICU-specific complications
Surgical (S)	Return to OR, operative time >6 hours, post-op intubation → surgical complications, procedural failures
Perinatal (P)	Apgar <7 at 5 min, birth trauma, NICU transfer → labor and delivery adverse events
ED (E)	Return to ED within 48 hours → premature discharge, missed diagnosis

Each trigger does not confirm harm, it identifies a case that warrants clinical review. The methodology was designed for a 20-record monthly sample reviewed manually by trained nurses. It was never designed to scale to every patient. AI changes that constraint.

AI Application: An automated implementation of the trigger tool runs nightly across all discharges. When a trigger fires, the system cross-references related clinical documentation to assess whether the trigger represents a likely adverse event.

Data that shows medication-associated kidney injury clustering in elderly surgical patients become institutional quality targets..

Classen, D. C., Resar, R., Griffin, F., Federico, F., Frankel, T., Kimmel, N., Whittington, J. S., Lichtner, A., & James, B. C. (2011). 'Global trigger tool' shows that adverse events in hospitals may be ten times greater than previously measured. *Health Affairs*, 30(4), 581–589.

<https://doi.org/10.1377/hlthaff.2011.0190>

Root Cause Analysis

Understanding of Need

When a serious patient safety event occurs, a wrong-site surgery, a fatal medication error, a patient death from an unrecognized deterioration, healthcare organizations are required to conduct a thorough investigation to understand what went wrong and prevent similar events in the future. The Joint Commission, which accredits most American hospitals, mandates that these "sentinel events" trigger a root cause analysis within strict timeframes: the investigation must begin within 72 hours and be completed within 45 days.

Root cause analysis is a systematic approach to understanding not just what happened, but why it happened. The goal is to look beyond the individual who made the final error and understand the system failures, communication breakdowns, and process gaps that allowed the error to occur.

The challenge is that the same clinical event is documented differently by different staff members. Reconstructing a coherent timeline from these fragmented accounts is extraordinarily labor-intensive. Investigators must read through pages of nursing notes, physician documentation, respiratory therapy records, pharmacy records, and safety reports. They must identify where the accounts align and where they conflict.

In the same way that Guideline Adherence can scale to include the entire patient population, it's valuable (completely impractical today) to run some version of an RCA on every harm event, not just sentinel events. In conversations with safety leaders, the concept of "mini-RCAs" are emerging as a promising approach to use AI to do an RCA of every event which massively reduces the burden on safety managers to review and triage reported events. AI does the initial RCA and uses reasoning to identify trends and focus review on only the highest value RCA results.

Joint Commission Root Cause Analysis: Chart Review Requirements

The categories below represent the questions that demand the most intensive clinical chart review, where AI-assisted timeline reconstruction and cross-source synthesis can most significantly accelerate the investigation.

RCA CATEGORY	QUESTION	SOURCES TO REVIEW	CHART REVIEW CHALLENGE
Communication	Were critical findings communicated between providers and acknowledged?	Nursing notes, physician notes, RT notes, handoff records	Escalations documented in one system may be absent from another
Assessment	Were changes in condition recognized and acted on in time?	Vital sign trends, nursing assessments, lab trajectories, scoring tools	Requires mapping deterioration against data available at the time
Information Availability	Did the decision-maker have access to critical results?	Lab acknowledgment logs, radiology reports, allergy records	Results can exist in the system but never be opened or referenced
Medication Use	Were medications ordered, dispensed, and monitored	CPOE orders, pharmacy logs, MAR, renal labs, alert	Requires tracing the full chain from order to administration

	correctly?	override logs	
Continuum of Care	Was information preserved across care transitions?	Transfer orders, handoff notes, discharge summaries	Information loss at handoff is a leading cause of sentinel events
Operative Care	Were safety checklists completed and intra-op events communicated?	Surgical checklists, anesthesia records, operative reports, recovery notes	Multiple providers document overlapping timeframes in separate systems
Staffing	Were staffing levels and competencies adequate for patient acuity?	Schedules, assignment sheets, acuity scores, overtime logs	Staffing data often lives in HR systems separate from clinical records
Patient Identification	Were ID protocols and consent verification followed at each step?	Consent forms, time-out documentation, wristband logs	Requires verifying every checkpoint in the identification chain

Source: Joint Commission Framework for Root Cause Analysis and Corrective Actions.

Proposed Approach

AI can dramatically accelerate the information-gathering phase of root cause analysis by extracting relevant details from across the medical record and synthesizing them into a unified timeline. The system reads documentation from nursing, physicians, respiratory therapy, pharmacy, and other sources, identifies details relevant to the event under investigation, and constructs a chronological reconstruction that makes gaps and discrepancies visible.

This does not replace the clinical judgment and systems thinking that quality professionals bring to root cause analysis but rather it compresses the time-consuming work of reading through records and reconstructing timelines.

Regulatory Alignment

This approach directly supports compliance with Joint Commission sentinel event requirements by accelerating the information-gathering process. Organizations can more reliably meet the 72-hour initiation requirement and 45-day completion deadline while conducting more thorough investigations.

Sentinel Event Review

Scenario: Two days into the hospitalization, a patient suffers a respiratory arrest requiring emergent intubation. He survives but requires an extended ICU stay. The safety team initiates a root cause analysis to understand what led to this preventable deterioration.

Clinical Context:

The investigator must piece together what happened from multiple documentation sources that tell overlapping but incomplete parts of the story. A nurse escalates a concern. A respiratory therapist responds with a recommendation. A physician writes a progress note that references neither. A critical lab result enters the system but is never acknowledged. By the time the rapid response is called, every piece needed to prevent the deterioration was present in the record but never assembled into a coherent picture.

Pt: Male, 74 | Encounter: ICU/Rapid Response

Nursing note 6:00 AM:working harder to breathe, using accessory muscles | requested RT evaluation

RT note 7:15 AM:Recommended BiPAP initiation, order placed, awaiting physician acknowledgment

Physician note 8:30 AM:will continue current O2 therapy and monitor closely (NO mention of BiPAP)

Blood gas 7:00 AM:pCO2 67, pH 7.28

Rapid response called: 9:45 AM,O2 sat 82%

AI Application: The AI ingests all documentation sources and constructs a unified timeline. It surfaces what no single clinician could see: a BiPAP recommendation never acted upon, a critical blood gas never acknowledged, and a two-hour window where intervention could have prevented the arrest. It maps these gaps to Joint Commission root cause categories automatically.

Sensitive Data Segmentation

Understanding of Need

Healthcare records contain some of the most personal information imaginable, and certain categories of health data require extra protection under the law. When someone seeks treatment for substance use disorder, receives reproductive healthcare, or accesses mental health services, that information carries legal protections beyond standard HIPAA privacy rules. The challenge for health information exchanges is that this sensitive information does not always announce itself with convenient labels or codes.

The recent (2024) introduction of TEFCA has drastically increased the volume of data exchange across state lines and will require scaled approaches to data segmentation, security and patient consent.

Federal and state regulations mandate that healthcare organizations segment certain types of sensitive health information before sharing it. The 2024 rule modifying 42 CFR Part 2 aligns substance use disorder confidentiality requirements with HIPAA, with full compliance required by February 2026. Meanwhile, California's AB352 requires segmentation of reproductive health data before sharing it across state lines. These regulations exist to protect patients from discrimination and to encourage people to seek care for stigmatized conditions without fear that their information will be inappropriately disclosed.

Modern clinical documents like C-CDAs contain both structured coded fields and extensive narrative sections where clinicians document the nuances of care. These documents frequently include Base64-encoded PDF attachments containing consultation notes, imaging reports, or historical records. Code-based filtering systems cannot read narrative text or decode and analyze PDF content, creating significant gaps in privacy protection.

Examples of Laws that protect data by type

Scope	Law	What Data Is Protected
Federal	42 CFR Part 2	Substance use disorder (SUD) treatment records from federally assisted programs. Protects any information that would identify a patient as having or having had an SUD.
Federal	45 CFR § 164.501 & § 164.508(a)(2)	Psychotherapy notes are defined as notes recorded (in any medium) by a mental health professional documenting or analyzing the contents of conversation during a private, group, joint, or family counseling session.
California	AB 352 (CMIA Civil Code §§56.101, 56.108, 56.110)	Sensitive services including abortion, contraception, and gender-affirming care. Mandates technical segregation of these records and prohibits automatic transmission to HIEs.
California	SB 81	Medical information and PHI including an individual's immigration status and place of birth.

Scope	Law	What Data Is Protected
Texas	SB 1188 (Health & Safety Code Ch. 183)	Electronic health records. Protects metabolic health and diet records. Prohibits storing credit scores or voter registration status within an EHR.
Texas	Health & Safety Code Ch. 81	Communicable disease records including HIV/AIDS test results, infection status, and partner notification information.
Oklahoma	Title 43A §1-109	All mental health and drug or alcohol abuse treatment information and communications between patients and licensed professionals.
Oklahoma	Title 63 §1-502.2	Communicable disease records. Protects HIV-related test results, infection status, illness, and exposure information.
Oklahoma	SB 1369 (63 O.S. §1-133 / OKSHINE)	Mandates provider reporting to state HIE but strictly excludes psychotherapy notes and 42 CFR Part 2 data from automatic transmission without explicit consent.

Proposed Approach

AI-powered segmentation addresses these gaps by identifying sensitive information based on what it actually means rather than what codes are present. The system reads and interprets narrative text the way a human reviewer would, understanding that a phrase like "discussed pregnancy options, patient elected termination" describes reproductive healthcare even without a specific procedure code attached.

Regulatory Alignment

This capability directly supports compliance with 42 CFR Part 2 as modified by the 2024 rule, ensuring that substance use disorder information receives appropriate protection. It also enables compliance with state-specific requirements like California's AB352 reproductive health privacy law. For organizations participating in the Trusted Exchange Framework and Common Agreement (TEFCA), comprehensive segmentation capability is essential for meeting the network's privacy and consent requirements.

AI Data Cop

Scenario: A California-based health information exchange receives a query from an out-of-state clinician requesting records for a patient who recently relocated. Under California law, reproductive health information requires segmentation before release to out-of-state requesters.

Clinical Context:

The patient's record includes a 47-page C-CDA document with several embedded PDF attachments. The structured problem list in the document contains only two conditions: hypertension and anxiety, neither of which triggers any privacy restrictions. However, one of the embedded PDFs is a consultation note from an OB/GYN visit about pregnancy options, including the patient's ultimate decision to have an abortion.

Pt: Female, 42 | Encounter: HIE Query/Out-of-State Request

Structured Problem List:hypertension, anxiety

Embedded PDF (OB/GYN Note):

Content:pregnancy options discussion, decision to have abortion

System Filter Result:Code-based filter: APPROVED for release (no codes trigger restrictions)

AI Analysis Result:REQUIRES SEGMENTATION per CA AB352 (reproductive health narrative)

AI Application: The AI system processes the complete document, including decoding and analyzing all embedded PDF attachments. Upon reading the OB/GYN consultation note, it recognizes the reproductive health content based on the meaning of the text.

Cautionary Framework: Responsible AI Deployment in Clinical Settings

Two failure modes pose significant risk if left unaddressed by federal policy. The first is operational: AI systems that overwhelm the humans they serve. The second is technical: AI systems whose reasoning cannot be trusted, explained, or reproduced.

I. Signal Overload

The IHI Global Trigger Tool demonstrated that adverse events occur in roughly one-third of hospital admissions, ten times the rate captured by voluntary reporting. When one multi-hospital system deployed electronic trigger tools, the volume was so overwhelming the organization disabled it entirely. The tool worked. The humans had no framework for processing the output.

Our team experienced this directly in sepsis detection work at emergency department triage. The model achieved strong performance (F1 0.71, published in JMIR). The modeling succeeded. The implementation failed. We did not account for clinical workflow and saved zero lives.

The principle: detection must be paired with prioritization, and prioritization with workflow integration. Systems should perform counterfactual analysis to surface the guideline failures most strongly associated with harm, transforming a firehose into a ranked, actionable list. Federal guidance should require AI safety tools demonstrate operational integration, not just detection accuracy.

II. Hallucination and Reasoning

LLMs produce hallucinations: clinically plausible statements that are factually wrong. Research documents hallucination rates of 1.47% and omission rates of 3.45% in clinical note generation. At health system scale, these translate directly to patient safety events.

The deeper issue is that LLMs reason by pattern completion, not clinical logic. Consider "last known well" in stroke care: establishing this timestamp requires synthesizing nursing notes, EMS reports, and family interviews, resolving contradictory accounts across multiple sources. An LLM might produce the correct answer, but there is no way to verify why, and no guarantee the same model produces the same answer after a version update. A wrong timestamp determines whether a patient receives tPA or is excluded from treatment.

Neurosymbolic reasoning as an alternative. Our approach, validated with the College of American Pathologists, decomposes clinical questions into logic trees of constrained sub-problems. Each node uses AI in a tightly scoped role. The tree composes results using explicit, inspectable logic, producing auditability (answers trace to source evidence), reproducibility (same inputs, same outputs regardless of model version), and targeted validation (each node tested independently).

Recommended Policy Guardrails

Require structured output over free-text generation for clinical determinations. Mandate hallucination and omission rate disclosure tested against clinician ground truth. Require reasoning transparency so every clinical determination has an auditable chain. Distinguish extraction from generation in federal

certification. Require post-deployment monitoring per FDA GMLP principles. Mandate implementation outcome disclosure, including cases where accurate detection failed to improve outcomes due to workflow barriers.

These recommendations align with FDA Good Machine Learning Practice principles and ASTP transparency requirements. We urge validation frameworks that hold vendors accountable for what they deliver in clinical practice, not just what their models achieve in a laboratory.

Conclusion

Healthcare quality improvement runs on chart abstraction and for twenty years, federal policy has pursued a straightforward goal: make clinical data machine-readable so this manual work becomes unnecessary.

Despite this investment, chart abstraction persists. Quality measure reporting alone costs U.S. physician practices \$15.4 billion annually. Health systems dedicate entire teams to registry abstraction - one academic medical center employs five full-time staff just to abstract surgical cases for a single registry. CMS leadership acknowledges that even basic structured data, such as blood pressure readings, often remain unreliable until organizations individually correct their workflows. The data standardization that was supposed to eliminate manual review never fully arrived.

The result is that quality improvement remains constrained by what organizations can afford to abstract. Registries limit their data elements. Quality measures simplify their specifications. Safety surveillance samples only a handful of records each month. Organizations discover guideline failures only after harm occurs because comprehensive monitoring would require reading every chart.

AI-based clinical natural language processing changes this calculus. Systems that achieve human-level accuracy while reducing abstraction time by 70–90% do not merely lower costs—they remove the constraint that has shaped what quality improvement can attempt.

When abstraction costs drop by an order of magnitude, quality measurement works the way it was always intended:

- Registries collect the clinical detail that drives improvement, extracted directly from operative notes and pathology reports, without requiring large abstraction teams
- Quality measures extend into narrative documentation where relevant clinical information actually resides, rather than contorting specifications to fit available structured fields
- Safety surveillance scales from monthly samples to continuous monitoring, surfacing adverse events that voluntary reporting misses
- Guideline adherence shifts from retrospective discovery of failures to prospective identification of care gaps while intervention is still possible
- Root cause analysis compresses from weeks of manual reconstruction to hours, enabling more thorough investigations that prevent recurrence

When the constraint lifts entirely, new possibilities emerge and applications that were never feasible when abstraction required human review:

- Quality measures too complex for manual abstraction become practical, such as the CAP pathology measures we validated at human-level accuracy after previous automation attempts failed
- Real-time guideline monitoring across entire patient populations, identifying the specific protocol failures most strongly associated with harm

- Provider-level performance measurement that enables gold-card qualification, replacing per-request prior authorization with demonstrated trust

This is not speculative. Our collaboration with the College of American Pathologists demonstrated AI abstraction matching experienced human abstractors on measures that had defeated prior automation. The capability exists. The question is how federal policy accelerates responsible adoption.

We urge caution alongside ambition. AI systems that overwhelm clinical workflows with unactionable alerts help no one. Hallucination rates that appear low in aggregate translate into patient safety risks at scale.

Pharos Health welcomes the opportunity to partner with HHS to make this transition responsible, validated, and real.

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EXHIBIT 1

Neurosymbolic AI for automated clinical quality measure
abstraction in pathology

CMSS 25

Neurosymbolic AI for automated clinical quality measure abstraction in pathology

Colleen Skau, PhD (College of American Pathologists)

Felix Brann (Pharos)

Alex Clarke, MD, PhD (Pharos)

Presenters

- **Colleen Skau, PhD** — Assistant Director, College of American Pathologists
- **Felix Brann** — CEO, Pharos
- **Alex Clarke, MD, PhD** — CMO, Pharos

Relevant Relationships

- Mr. Brann and Dr. Clarke serve as executive officers of Pharos, a company collaborating with the College of American Pathologists (CAP) and other professional societies to develop AI-enabled automated clinical abstraction tools.
- Dr. Skau has no disclosures.

1: Introduction

Extracting clinically relevant measures is challenging

2: A New Paradigm

Neurosymbolic AI for automated quality measure abstraction

3: Current Status

Development, evaluation and results

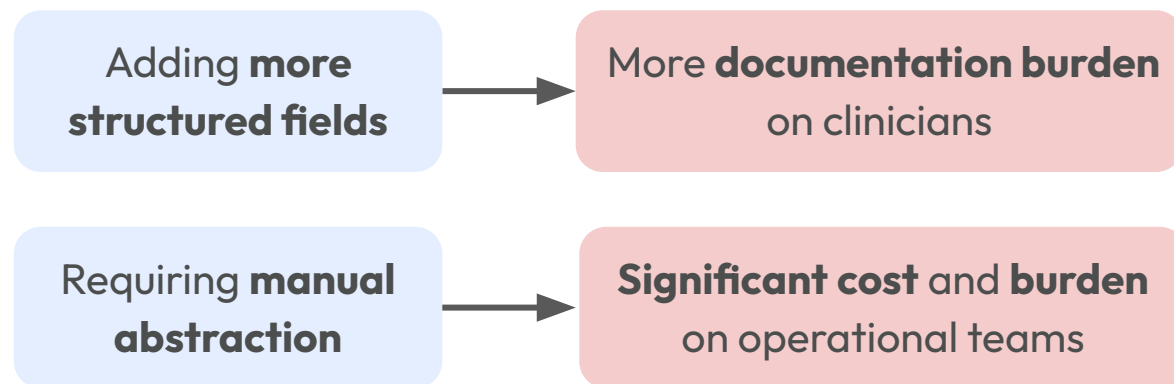
4: Next Steps

1: Introduction

Extracting clinically relevant measures
is challenging

The Goal: Capture clinically relevant measures while minimizing friction

- It's crucial to **minimize the burden of registry submission** for partner health systems
- Data is **useless unless it's accurate**



But... Over-simplifying measures to rely only on structured fields undermines the purpose of a clinical registry

- **Measures become clinically irrelevant** and fail to reward high-quality care
- **The dataset loses the richness needed** for nuanced research, risk adjustment, and subgroup analyses

The Challenge: Clinically relevant concepts live in messy narrative data

SPECIMEN SOURCE:

A. Urinary bladder

FINAL PATHOLOGIC DIAGNOSIS:

Urinary bladder, bladder tumor, poorly differentiated

Poorly differentiated malignancy, see comment.

Angiolymphatic invasion is present; muscle is present but not involved.

CLINICAL HISTORY: BPH, local w/urinary obstruct w/other low urinary tract sx, hydronephrosis bilateral, urinary retention. Unspecified hydronephrosis [ICD-10 : N13.30]

COMMENT: Tumor cells are very focally positive for Oscar, and positive for CD68. CD43 is negative. Ki-67 is 80%. The tumor morphology was compared to his previous biopsy and resection, with a similar appearance, and the biopsy had a more reassuring keratin profile, supporting the diagnosis. The following immunostains are negative: DESMIN, CD45, SMA, CAM5.2, AE1/3, ALK1, CK7, CK20, Gata3, S-100, NKX3.1, p40, CDX2, SOX10. Intradepartmental consultation was performed with concurrence.

GROSS DESCRIPTION

A. Bladder tumor

Received in formalin labeled with proper patient identification and "bladder tumor". The specimen consists multiple pieces of tan friable tissue measuring 7.0 x 6.5 x 2.5 cm in aggregate. The specimen is filtered through a biopsy bag and totally submitted in 12 cassettes labeled A1-A12.

Measure Description

Percentage of urinary bladder carcinoma pathology reports that include:

Procedure

- Histologic tumor grade
- Histologic type
- Presence/absence of muscularis propria
- Presence/absence of **lymphovascular invasion**
- Tumor extent

The Challenge: Clinically relevant concepts require deep contextual understanding

CPT Code(s): 88305x4

Laboratory Provided ICD & SNOMED Code(s): C44.41, C44.519, D04.62, L57.0 238528004, 402509003, 92352002, 403878007, 201101007, 254701007, 92380000, 254656002

SPECIMEN SOURCE:

RIGHT FOREHEAD

RIGHT ANTERIOR NECK LINE

LEFT MID STERNUM

LEFT FOREARM

FINAL PATHOLOGIC DIAGNOSIS:

A. Skin, right forehead, shave biopsy: Actinic keratosis.

B. Skin, right anterior neck line, shave biopsy: Basal cell carcinoma, nodular type; - **present at peripheral and deep margins.**

C. Skin, left mid sternum, shave biopsy: Basal cell carcinoma, superficial type; - **present at peripheral margin.**

D. Skin, left forearm, shave biopsy: Squamous cell carcinoma in situ; - **present at peripheral and deep margins.**

This case has been reviewed and dictated by
M.D., Ph.D., Board Certified Dermatopathologist.

Measure Description

Percentage of final pathology reports for basal cell carcinoma of the skin that include:

- Histologic subtype
- **Margin status**
- Presence or absence of invasion beyond the reticular dermis (or for Mohs specimens, anatomic level of invasion)
- Perineural invasion where required

Existing solutions fall short

Human abstraction

- Expensive
- Time-consuming
- Requires validation and training

Keyword based data extraction

- “thyroid, right isthmus nodule” would miss a nodule on the left side of the thymus
- Intractable to develop a complete set of discrete keywords
- **High rate of false negatives**

Rules based data extraction

- “*bronc* biops*” would identify “bronchial biopsy”, “bronchus, biopsy”, and “bronchial biopsies”
- Rules are complex to develop and brittle in practice
- **High rate of false positives**

Existing solutions fall short... sometimes catastrophically



Rule	Data Received
GE* biops*	The patient is “ , George”; biopsy, left ear

Some valuable measures are completely intractable with current solutions

SPECIMEN SOURCE:

- A. GASTRIC BIOPSY
- B. MID ESOPHAGEAL BIOPSY

FINAL PATHOLOGIC DIAGNOSIS:

A. Stomach, biopsy:

- Intestinal metaplasia (complete-type), involving corpus/fundic mucosa.
- Pathologic evidence of Helicobacter infection is not identified.
- See comment.

B. Esophagus, mid, biopsy:

- Squamous mucosa with no significant histopathologic change.
- Pathologic evidence of eosinophilic esophagitis is not identified.

Measure Description

Percentage of pathology reports for **gastric specimens** with intestinal metaplasia that comment on:

- Presence or absence of H. pylori
- Presence or absence of dysplasia and if dysplasia is present, grade of dysplasia
- If dysplasia is absent, completeness of metaplasia

The College of American Pathologists had to abandon automatic collection of this measure.

2: A New Paradigm

Neurosymbolic AI for automated quality
measure abstraction

Modern AI gives us tools to tackle this problem – but a naive approach comes with significant risks

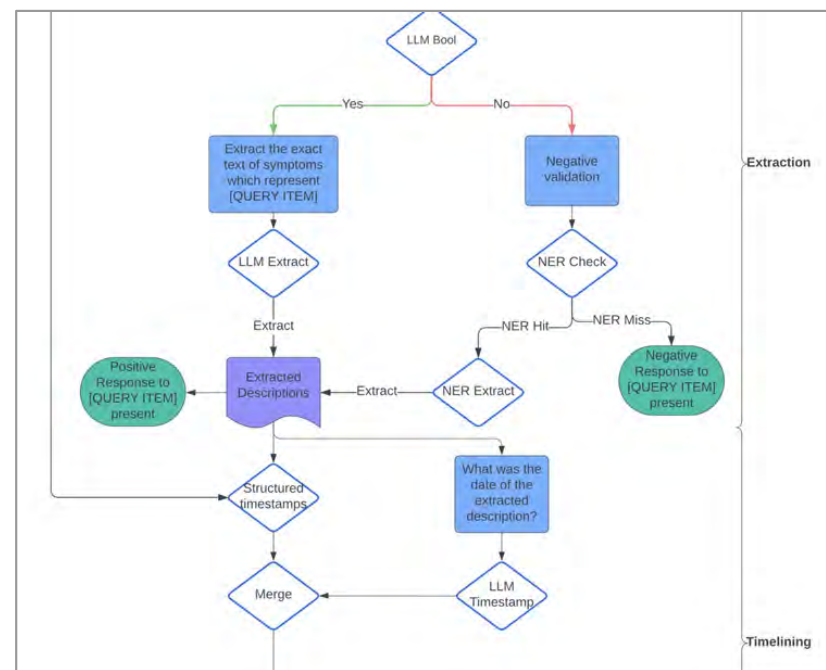
- Models directly **hallucinate facts from the medical record**
- Models use ‘common sense definitions’, **instead of the precise definitions laid out in measures**
- Model decision-making is **opaque and arbitrary**

The result:

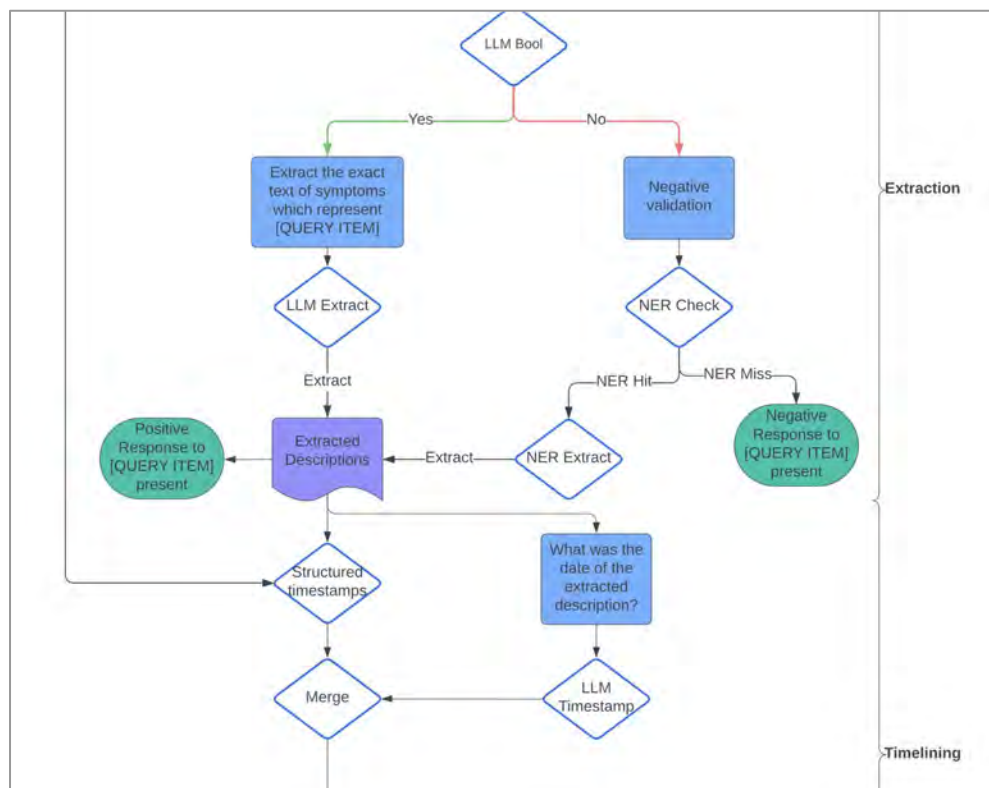
**Insufficient accuracy
and high variance**

Neurosymbolic AI breaks down the problem and composes many different models together to solve it

COLLEGE of AMERICAN PATHOLOGISTS		CAP QCDR Measure SCC Reporting
	- C44.722 Squamous cell carcinoma of skin of right lower limb, including hip - C44.729 Squamous cell carcinoma of skin of left lower limb, including hip - C44.82 Squamous cell carcinoma of overlapping sites of skin - C44.92 Squamous cell carcinoma of skin, unspecified	
Denominator Exclusions	1. Specimens that are exclusively in situ disease (ICD10 D04.0, D04.10, D04.111, D04.112, D04.121, D04.122, D04.20, D04.21, D04.22, D04.30, D04.39, D04.4, D04.5, D04.60, D04.61, D04.62, D04.70, D04.71, D04.72, D04.8, D04.9)	
Denominator Exceptions	- Biopsy or cytology specimens - Specimens displaying fragmentation - No residual carcinoma - Necrotic tissue - Specimen site other than cutaneous or cancer other than squamous cell - Tumor is less than 2 cm in diameter	
Numerator Statement	Pathology report contains a statement on all of the following: - Margin status - Degree of differentiation/histologic grade - Depth of invasion or for Mohs' excisions level of invasion	

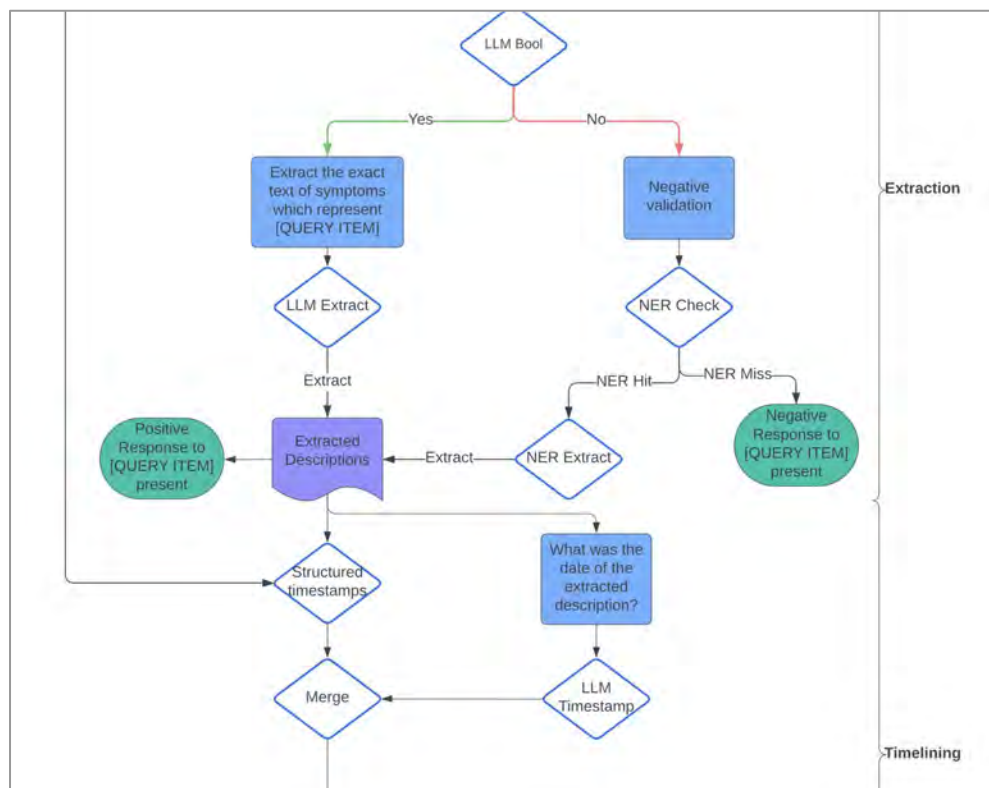


Neurosymbolic AI breaks down the problem and composes many different models together to solve it...



- Measure definitions are broken down into **small sub-problems**
 - How many samples are in the report?
 - Is a lesion *in situ* only?
 - Is a grading statement required?
- **Custom models** are used to solve each sub-problem, and the solutions are composed together

... letting us leverage powerful models, with tight constraints and auditable reasoning



- Use of AI is tightly scoped, **minimizing hallucination** and forcing models to **use precise definitions**
- Evaluation is highly consistent, resulting in **low-variability outputs**
- Produces **fully auditable abstraction reasoning and confidence** for every decision

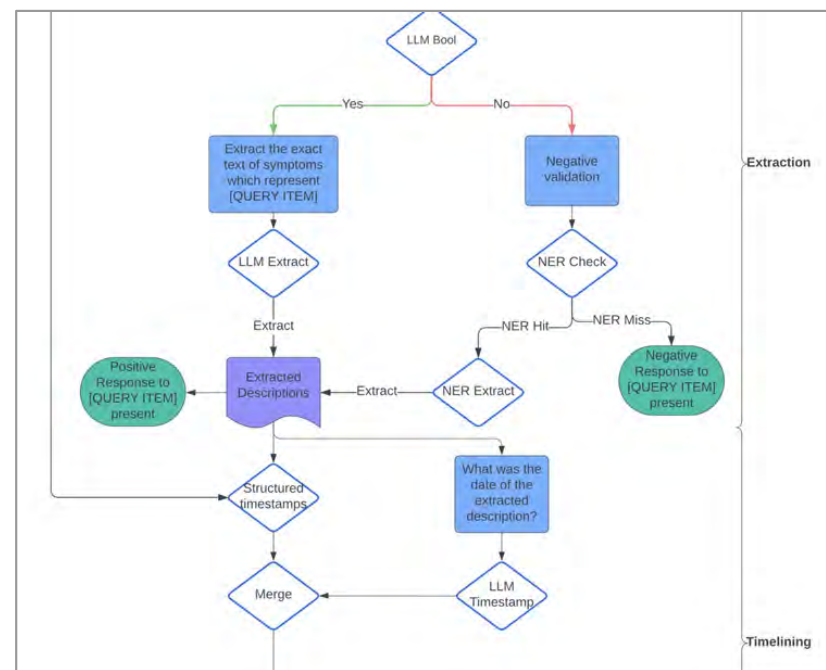
Supporting evidence and reasoning

3: Current Status

Development, evaluation and results

Initial Neurosymbolic trees are generated from existing manual abstraction instructions

 COLLEGE of AMERICAN PATHOLOGISTS	
CAP QCDR Measure SCC Reporting	
	• C44.722 Squamous cell carcinoma of skin of right lower limb, including hip • C44.729 Squamous cell carcinoma of skin of left lower limb, including hip • C44.82 Squamous cell carcinoma of overlapping sites of skin • C44.92 Squamous cell carcinoma of skin, unspecified
Denominator Exclusions	1. Specimens that are exclusively in situ disease (ICD10 D04.0, D04.10, D04.111, D04.112, D04.121, D04.122, D04.20, D04.21, D04.22, D04.30, D04.39, D04.4, D04.5, D04.60, D04.61, D04.62, D04.70, D04.71, D04.72, D04.8, D04.9)
Denominator Exceptions	1. Biopsy or cytology specimens 2. Specimens displaying fragmentation 3. No residual carcinoma 4. Necrotic tissue 5. Specimen site other than cutaneous or cancer other than squamous cell 6. Tumor is less than 2 cm in diameter
Numerator Statement	Pathology report contains a statement on all of the following: - Margin status - Degree of differentiation/histologic grade - Depth of invasion or for Mohs' excisions level of invasion



Neurosymbolic trees are refined against real and synthetic patient examples...

Edge cases are reviewed by clinicians and clarifications iteratively improve trees

! Case needs review

...with edge cases automatically identified and added to the measure definition and model tree

Pharos

Alex Clarke | Clinical Reviewer

Menu < Question Library > Create a new question

Save question

Abstraction question definition Case-based refinement

Peripheral Margin Status Included

Question

Does the pathology report include a statement about the peripheral margin status for the invasive malignant cutaneous melanoma specimen reviewed in this report?

Detailed Specification

This field assesses whether the pathology report includes a statement about whether invasive malignant melanoma extends to the peripheral (lateral) margin of the excised specimen. Evaluation of the peripheral margin is essential for determining the completeness of excision and planning further treatment.

The report must clearly state, or reasonably imply, whether the peripheral margin is involved or uninvolved by melanoma. Terminology may include "peripheral margin," "lateral," or equivalent phrases. If multiple invasive melanoma specimens are present, each must include a statement about peripheral margin status to answer "Yes."

Note: A statement that the invasive melanoma is completely surrounded by melanoma specimen (i.e., no invasive tumor at any inked edge) will be accepted as document (lateral) margin status and scored as peripheral margin uninvolved for the invasive component, even if melanoma in situ reaches a peripheral margin—unless the report elsewhere limits the margin comment exclusively to the deep/base margin.

Accept A

Reject R

Results:

24 cases require your clinical judgment to refine the measure. Review these cases to help the AI understand edge cases and improve accuracy.

Estimated time: 10-15 minutes

Pattern detected

You marked 4 of 4 reports where the invasive melanoma was stated to be completely surrounded by *in situ* disease as inclusion cases.

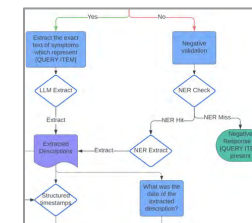
Would you like to explicitly add a note that statements that the invasive melanoma is completely surrounded by *in situ* disease are to be counted as peripheral margin inclusions?

Yes please

Generating measure definition...

V3 created

Type a message



Edge cases are reviewed by clinicians and clarifications iteratively improve trees

⚠ Case needs review

Model evaluation process

- Evaluation was split over 4 different CAP measures. Reports were assigned to each using diagnosis codes.
- 500 pathology reports randomly subsampled from each measure from a feed of 100k reports – giving 2,000 reports.
- All reports were **independently double-abstracted** by experienced third-party abstractors trained on each measure, giving a human vs human IRR.
- Final decisions were made on each report by third-party RN abstraction managers producing a gold-label.
- The neurosymbolic models were then run on the same reports and compared to the gold standard.

The result: Human-level abstraction performance

Standard Gold	Denominator Exclusion	466	9	0	1
	Numerator Pass	5	754	8	0
	Numerator Fail	2	4	353	4
	Numerator Exception	0	0	3	9
		Denominator Exclusion	Numerator Pass	Numerator Fail	Numerator Exception
		Model Predicted			

	IRR	Accuracy
Human	0.918	95.4%
AI	0.922	95.7%

No statistical difference between human and AI abstractors

4: Next Steps

Project evolution

Advantages of human-level automated abstraction

New measures can be clinically complex without adding burden

- Complex measure data can be collected without requiring costly abstraction
- Measures can be created to reflect new clinical practice guidelines without adding documentation burden

Reduced friction and increased value for partner practices

- Practices don't have to filter data before sending it to the registry
- Vastly more accurate measurement means dramatically fewer case corrections

Hugely accelerated measure development loop

- New measures can be applied instantly to a corpus of pathology reports.
- AI surfaces edge cases for review by measure developer to allow measure refinement before deployment

Next steps for this project

Adding new measures

- Continuing to train and validate model on additional complex measures
- Learnings from new measures provide opportunities to improve existing measures

Expanding the training/testing set

- Different clinicians use different phrases
- Different LISs capture data differently—if you've seen one LIS, you've seen one LIS

Developing deployment pathway

- Goal is to enable the College to deploy Pharos abstraction to collect data for new, clinically interesting measures

Results will be published!

What else can AI do for us?

Current pathology measures are limited: process measures focusing on documentation

- Partly a limitation due to the complexity of narrative reports
- PRO/PROMs are not possible for pathology (right now)
- What about outcome/intermediate outcome measures?

More complex measures that would be a burden on a practice

- Example: “diagnostic uncertainty” measurement—can we quantify how often a practice uses phrases such as “consistent with” or “suspicious for”? Stratify by specimen type? By anatomic location?

Beyond measurement: AI to help practices identify areas of improvement

- Subtle opportunities
- Turnaround time for specific specimen types, e.g. placental specimens
- Clinical practice guideline failures not present in measures

Other use cases: Automated detection of clinical practice guideline deviation

Day 1

Hospital Admission
Patient admitted, preoperative labs reviewed, VTE prophylaxis initiated, and anesthesia evaluation completed.

Guideline 5.2 · VTE · Admission

Perform standardized VTE risk assessment upon admission and document the VTE risk in the patient record.

✓ Completed Show evidence

Guideline 5.6 · VTE · Admission

If patient is at assessed at moderate/high VTE risk, initiate a chemoprophylaxis agent unless contraindicated.

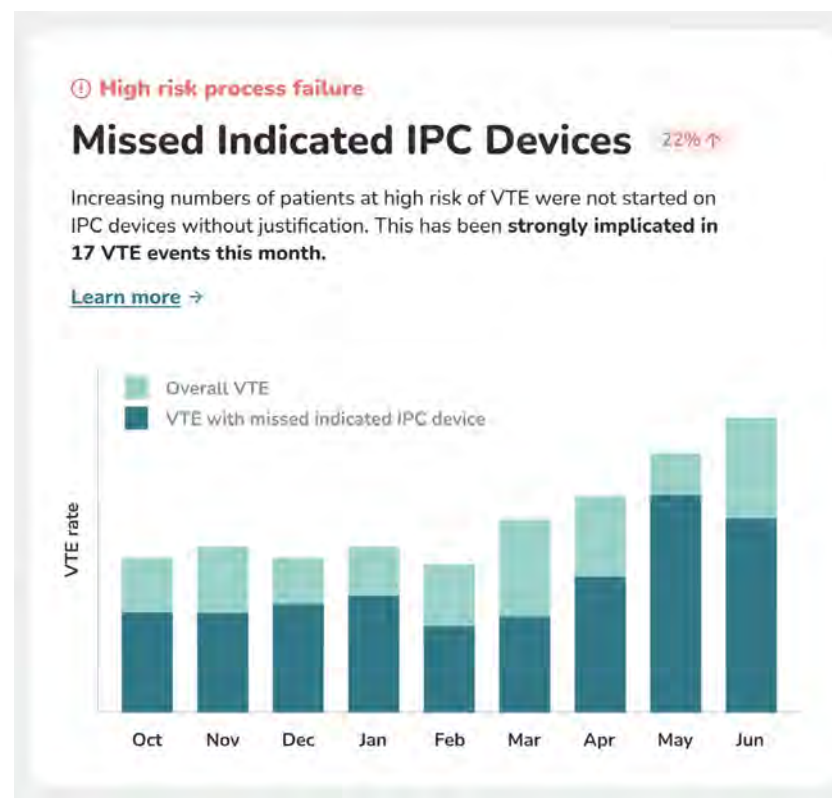
✓ Completed Show evidence

Guideline 5.3B · VTE · Admission

Patients assessed as being high risk for VTEs should be started on an intermittent pneumatic compression (IPC) device unless contraindicated.

✚ The patient was assessed to be in the high risk category for VTEs, as documented in the VTE assessment document completed on admission and referenced in the **admission note completed 02/12/24**. However, the **nursing record completed 11:05am on 02/12/24** indicates that the patient was instead given compression stockings.

❗ Missed Hide evidence



Other use cases: Automated detection & analysis of safety events

Pharos

EncountersHarm Event InsightsGlobal QualityQI Projects

Ellen Howard
Quality Program Manager

Patient Encounters

RCA Inbox

Filter by

All safety events

All triggering events

Last 30 days

Sort by

Newest

Search

#	Patient Name	MRN	Primary Safety Event Type	Triggering Event	Status
229	Clara Brown	34211876	Medication or Other Substances	Staff safety report	Awaiting RCA
228	Emily Rodriguez	59023145	Surgery, including invasive procedures	Auto-detected safety event	Safety event reported
227	Erica Baldwin	78065439	Healthcare-associated Infection	Patient complaint	Awaiting RCA
226	Danielle Ruiz	11678324	Perinatal	Staff safety report	Safety event reported
225	Noah Ellison	90456287	Device or Medical/Surgical Supply	Auto-detected safety event	Safety event reported
224	Maya Benson	22347810	Blood or Blood Product	Auto-detected safety event	RCA in progress
223	Elliot Crenshaw	67509133	Surgery, including invasive procedures	Staff safety report	Closed without RCA
222	Natalie Greer	83402966	Pressure Injury	Staff safety report	RCA in progress
221	John Thompson	47281936	Fall	Auto-detected safety event	Closed without RCA
220	Linda Martinez	58327412	Fall	Patient complaint	RCA complete
219	Robert Chen	69158203	Healthcare-associated Infection	Staff safety report	Closed without RCA

Pharos

EncountersHarm Event InsightsGlobal QualityQI Projects

Ellen Howard
Quality Program Manager

February 14, 2025

21:00 (14 Feb)

Patient Clara Brown arrived with infective exacerbation of COPD.

EMR Evidence

21:45 (14 Feb)

Oxygen saturation dropped below 88% with VBG showing PCO2 of 67 mmHg.

EMR Evidence

21:50 (14 Feb)

Jane Smith RN recommended BiPAP but Dr. Evans declined.

EMR Evidence

Key failure: Missed Guideline #2.3.1

Non-invasive ventilation was not initiated within 30 minutes of identifying hypercapnia (PCO2 67 mmHg)

Respiratory Support ProtocolEMR: Jane Smith RN

Key failure: Missed Guideline #4.1.2

There is no documentation that Dr. Evans implemented team-based decision making, ignoring both nurse and RT recommendations

Multidisciplinary Care StandardEMR: John Brown

EMR Evidence

Emergency Department: Nursing Note

Jane Smith RN | 14 Feb 2025, 21:50

URGENT PATIENT: Clara Brown
DOB: 05/12/1962
MRN: rca-2
ENCOUNTER: ED-20250214-001

URGENT RESPIRATORY STATUS UPDATE (21:50):
Patient's respiratory status has continued to deteriorate despite interventions. Oxygen saturation dropped to 88% on 4L nasal cannula and has not improved with increase to 6L/min. Patient now with increased work of breathing, using accessory muscles prominently, and showing signs of fatigue.

VITAL SIGNS (21:50):
Temperature: 101.0°F Heart Rate: 122 bpm (increased from previous) Blood Pressure: 142/88 mmHg Respiratory Rate: 32 breaths per minute O2 Saturation: 87% on 6L/min oxygen

ASSESSMENT:
Patient appears increasingly distressed with labored breathing. Venous blood gas results show PCO2 of 67, indicating significant hypercapnia. Breath sounds remain diminished with prominent wheezing and rhonchi bilaterally. Patient becoming anxious and stating "I can't catch my breath."

COMMUNICATION:
Dr. Evans immediately notified of:
1. Continued oxygen desaturation despite increased flow
2. VBG results showing PCO2 of 67
3. Patient's increasing respiratory distress.
Suggested BiPAP initiation based on

Acknowledgements

Sarah O'Connor

former Assistant Director, Pathologists Quality Registry

Dr Gregory Bocsi, DO, FCAP

Chair, Value-Based Payment and Healthcare Policy Committee, CAP

Dr Thomas Cooper, MD, FCAP

Member, Value-Based Payment and Healthcare Policy Committee, CAP

The Team at Kiser Healthcare Solutions

Questions?

Speak with us at Table 10